M1 Stress Fragility Analysis: Review of Seismic Tool Worked Example 91 and Proposed Corrections

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Abstract

This report reviews the Monte Carlo M1 stress fragility analysis of Seismic Tool Worked Example 91 (WE91), as described in GMT-DOC-06175 [1] and implemented in `post_process_s91_stress.m` and `earthquake_MonteCarlo.m`. It finds three defects in the Monte Carlo implementation and five in the method and its inputs, and proposes a correction for each, with its derivation and justification. The implementation defects are: (i) the number of earthquakes per return-period bin is overestimated by a factor of 4.3; (ii) the first earthquake of every 50-year lifetime is discarded; (iii) earthquakes rarer than $2{,}512$ years are never simulated. Defects (i) and (ii) partly cancel, so they must be corrected together. The method and input defects are: (iv) the conditional-mean-spectrum (CMS) record used as the mean of each stress distribution responds about twice as strongly as the mean of the 16 conditional-spectra (CS) records at the survival level; (v) a single 50 psi gravity baseline is applied to every stress group; (vi) the maximum over segments is used in place of a system failure probability, with one Weibull law for all surface finishes; (vii) the "95th-percentile" hazard factor of 1.3 is not traceable to the site hazard study; (viii) the stress inputs predate the 2025 stress model and cover one telescope orientation. The Weibull constants K and β of the analysis, flagged as untraceable in [2, 1], are derived here from the glass test data and reproduce all three 50-year stress limits used in the code. A revised algorithm, a verification plan and a prioritised implementation plan are given; the first four steps need no new simulations.

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1 Introduction

1.1 Scope

The M1 stress fragility analysis estimates the probability that a primary-mirror segment develops a crack during a 50-year observatory lifetime. It accounts for the cumulative, history-dependent damage done by gravity, thermal and annealing stress, and by the random sequence of earthquakes expected at the site. WE91 was delivered at the January 2023 Seismic Tool Workshop [2] as a worked example, not a complete analysis. GMT-DOC-06175 [1] documents its implementation. The comments D. Schwartz added to that document are taken into account here.

The material reviewed is:

- GMT-DOC-06175, *M1 Stress Fragility Analysis*, v1.0 draft, including the comment thread [1];
- `fragility/post_process_s91_stress.m` and `fragility/earthquake_MonteCarlo.m`;
- the WE91 run matrix and summary output (`GMT_Earthquakes_S91_.xlsx`, `S91res.mat`) and the run driver `Run_GMT_Sims_2020_S91.m`;
- `stress_post_local_7_S91_WorkedExample3.m`, the stress post-processing script;
- the Seismic Tool Workshop draft [2], the SPIE paper [3], AECOM Addendum 2 [4], the M1 damper peer-review slides [5], and the seismic analysis handoff memo [6].

1.2 Limitations

The stress time series of the 42 WE91 runs (`T:\SeismicResults\S91`) were not available for this review. The effect of each correction is therefore quantified with a Python port of `earthquake_MonteCarlo.m`, fed with synthetic equivalent-stress statistics sized to match the histograms in [1]. These numbers show the direction and approximate size of each effect, not the corrected WE91 result. The port contains the original loop, the loop of GMT-DOC-06175 Fig. 9, and the corrected implementation (`mc.py`, `hz.py`, `study.py`). With the S91 stress files, the same scripts produce the corrected figures.

1.3 Summary of findings

Table 1 lists the findings. Items C1–C4 concern the implementation and items M1–M5 the method and its inputs.

Table 1: Summary of findings and proposed corrections.

ID	Finding	Correction	New runs
C1	Events per bin computed from the exceedance rate at the bin centre ($\times 4.3$ too many).	Use the difference of exceedance rates at the bin edges, or sample Sa directly from the hazard curve (Section 3.1).	No
C2	First earthquake of each lifetime discarded; empty lifetimes mis-indexed.	Assign events to lifetimes by $\lfloor t/50 \rfloor$ and accumulate (Section 3.2).	No
C3	Events rarer than 2,512 yr never simulated; SLE CMS $\times 1.36$ and $\times 1.85$ unused.	Extend the intensity range to the hazard data limit; check convergence of the upper bound (Section 3.3).	Partly
C4	No random seed; safety-factor sweep overwritten; plotting and labelling errors.	Housekeeping items listed in Section 3.4.	No
M1	SLE CMS response about $2\times$ the CS mean; the CMS lies outside the CS ensemble.	Diagnose the Sa(0.1 s) anchoring; take the level from the CS and the trend from the CMS; fit a lognormal; move to multiple-stripe analysis (Section 4.1).	Eventually
M2	One 50 psi gravity baseline for all groups (vent hole is 100–111 psi).	Per-segment, per-group baseline, zenith-weighted in equivalent time (Section 4.2).	No
M3	Maximum over segments instead of a system probability; Fibrefrax Weibull law for every finish.	Sum of Weibull risks of rupture with per-group area and finish; record bootstrap (Section 4.3).	No
M4	1.3 fractile factor not traceable; mislabelled.	AECOM 95th/mean ratios now; full fractile curves later (Section 4.4).	No
M5	Superseded stress model; single orientation (az $0°$, ZA $60°$).	`Stress_upd_2025`; zenith and azimuth treated as aleatory variables (Section 4.5).	Yes

2 The method as implemented

2.1 Equivalent time and equivalent stress

Sub-critical crack growth in glass makes the time to failure under a constant stress s scale as s^{-n}, where n is the stress-corrosion exponent. Two stress histories therefore produce the same damage when their integrals of $s^n\,dt$ are equal. This is the basis of the *equivalent time* at a reference stress s_r:

$$t_{\mathrm{eq}} = \sum_i \Delta t \left(\frac{s_i}{s_r} \right)^n , \tag{1}$$

where s_i is the total stress at sample i and $\Delta t = 0.01$ s. Equivalent times from successive loadings are additive. The *50-year equivalent stress* is the constant stress that produces the same damage in $t_{50} = 50 \times 365 \times 24 \times 3600$ s:

$$s_{\mathrm{eq}} = s_r \left(\frac{t_{\mathrm{eq}}}{t_{50}} \right)^{1/n} . \tag{2}$$

The analysis uses $n = 30.5$. The reference stress s_r cancels in every final result.

2.2 Failure probability

The probability of crack formation after an accumulated equivalent time T is

$$F(T) = 1 - \exp\Big[-(K\, s_r^{\,n}\, T)^{\beta}\Big], \tag{3}$$

with $K = 3 \times 10^{-101}$ and $\beta = 0.4522$ (stress in psi, time in seconds). Section 4.3.1 derives these constants.

2.3 Monte Carlo over lifetimes

For each stress group, `post_process_s91_stress.m` computes s_{eq} for each of the 42 runs (Table 2), after multiplying the mechanical stress by a safety factor SF and adding 200 psi of thermal and annealing stress. It then takes the maximum over the seven segments. A Gaussian is assigned to each CMS scale factor: its mean is the CMS run and its standard deviation is the CMS run times the coefficient of variation of the 16 CS runs at the same level. Each scale factor is mapped to a return period through the $T = 0.1$ s mean hazard curve.

`earthquake_MonteCarlo.m` then simulates $N_{\mathrm{sim}} = 5000$ lifetimes. It scatters earthquakes uniformly in time over $N_Y = 50 N_{\mathrm{sim}}$ years, draws an equivalent stress for each, converts it to equivalent time with Equation (1), and accumulates it on top of the baseline equivalent time

$$T_b = t_{50} \left(\frac{s_b}{s_r}\right)^{n}, \qquad s_b = 200 + SF \times 50 \text{ psi}. \tag{4}$$

The expected failure probability is the mean over lifetimes, $\bar{F} = N_{\mathrm{sim}}^{-1} \sum_{\ell} F(T_\ell)$. It is reported as the 50-year equivalent stress that would give the same probability:

$$s_{\mathrm{eq}}^{\mathrm{MC}} = \left[\frac{\big(-\ln(1-\bar{F})\big)^{1/\beta}}{K\, t_{50}}\right]^{1/n}. \tag{5}$$

Because $F \propto s^{n\beta}$ for small F and $n\beta \approx 13.8$, an error by a factor r in $\bar{F}$ changes $s_{\mathrm{eq}}^{\mathrm{MC}}$ by only $r^{1/13.8}$: a factor of 4 in probability is about 10% in stress. All comparisons below report both quantities.

Table 2: WE91 run matrix (from `GMT_Earthquakes_S91_.xlsx`). All runs at azimuth $0°$, zenith angle $60°$. The comments on lines 37–38 of `post_process_s91_stress.m` label runs 33–37 as OLE; the run matrix shows they are SLE, which is how the code uses them.

Runs	Records	Level	Scale factors
1–16	`SLE_short_2019_1..16` (CS)	SLE (2475 yr)	1
17–32	`OLE_short_2019_1..16` (CS)	OLE (100 yr)	1
33–37	`SLE_short_2018_1` (CMS)	SLE	0.54, 0.735, 1, 1.36, 1.85
38–42	`OLE_short_2018_1` (CMS)	OLE	0.54, 0.735, 1, 1.36, 1.85

3 Corrections to the Monte Carlo implementation

3.1 C1 — Number of earthquakes per bin

Finding. Line 38 of `earthquake_MonteCarlo.m` sets the number of earthquakes in bin k to

```
numE = round(nYT./rp);  % rp = 10.^(bin centre)
```

that is, $N_Y/\bar{T}_k$, where $\bar{T}_k$ is the return period at the bin centre. This is the expected number of earthquakes whose intensity *equals or exceeds* the bin-centre intensity, not the number whose intensity falls *inside* the bin.

Derivation. Let $\lambda(a)$ be the annual rate of exceeding spectral acceleration a, so that the return period is $1/\lambda$. Earthquake occurrences are modelled as a Poisson process. The expected number of events with intensity in $[a_k, a_{k+1})$ over N_Y years is

$$\mathbb{E}[N_k] = N_Y\,[\lambda(a_k) - \lambda(a_{k+1})] = N_Y\left[10^{-x_k} - 10^{-x_{k+1}}\right], \tag{6}$$

where $x_k = \log_{10}(1/\lambda(a_k))$ are the bin edges in log return period. With uniform bins of width Δ and centre c_k, Equation (6) becomes $N_Y\,10^{-c_k}\,(10^{\Delta/2} - 10^{-\Delta/2})$. The code's count is therefore too high by

$$\frac{1}{10^{\Delta/2} - 10^{-\Delta/2}} = 4.34 \quad \text{for } \Delta = 0.1. \tag{7}$$

The error grows without bound as the bins are refined, which is a clear sign that the count is wrong. For WE91 ($N_Y = 250{,}000$, edges 1.7:0.1:3.4) the code generates 21,187 earthquakes where 4,889 are expected (Figure 1): 4.2 per lifetime instead of 0.98. The method described in the workshop document (§7.7 of [2]) and in the SPIE paper [3] is the correct one, Equation (6); it was explained there with the example "55 earthquakes of 9-year return period or greater minus 45 of 11-year or greater gives 10 in the bin". The implementation does not follow it. The commented alternative on lines 36–37 (`nYT./diff(10.^rp_bins)`) is also incorrect.

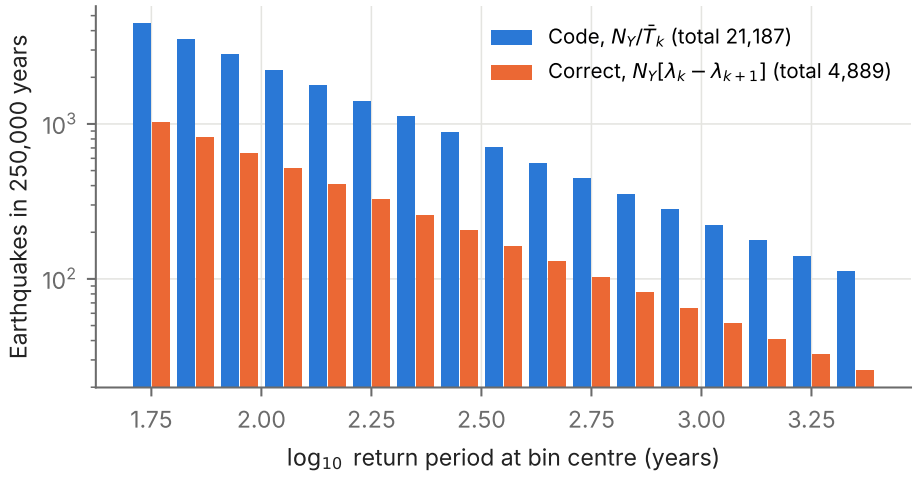


Figure 1: Earthquakes per return-period bin over 250,000 years for the WE91 bins: code (`round(nYT./rp)`) versus the correct count, Equation (6).

Correction. Replace the deterministic count with a Poisson draw of mean Equation (6):

```
lam  = 10.^(-rp_bins);                % exceedance rate at bin edges
numE = poissrnd(nYT * (lam(1:end-1) - lam(2:end)));
```

The Poisson draw is preferred to `round` because it keeps the variability of the number of events, which matters for lifetimes containing a rare earthquake. A better alternative removes the bins altogether (Section 3.3).

3.2 C2 — Lifetime bookkeeping

Finding. The accumulation loop of `earthquake_MonteCarlo.m` is

```
81  for k = 1:length(earthquake_time)
82      if (earthquake_time(k) < (kk*Nyears))
83          stress_eq = stress_mean(eqid(k)) + randn(1)*stress_std(eqid(k));
84          teT = teT + (stress_eq/sigmaC).^n .* t1_50year;
```

```
85      else
86          Ft(kk) = 1-exp(-(K*sigmaC.^n.*teT).^beta);
87          Te(kk) = teT;
88          kk = kk+1;
89          teT = teT_baseline;     % event k is never added
90      end
91  end
```

The event that crosses a lifetime boundary closes the previous lifetime and is then discarded. With the overcounted events of C1 this removes about $5{,}000$ of $21{,}187$ events. GMT-DOC-06175 Fig. 9 moves the accumulation outside the `if`, which fixes this. Both versions use `if` where a `while` is needed: when a lifetime contains no earthquake, the lifetime index falls behind by one and the next events are attributed to the wrong lifetime until it catches up. Lifetimes left without an index keep the `zeros` initial value of `Ft` instead of the baseline probability $F(T_b)$. With the corrected counts of C1, about 37% of lifetimes are empty.

The Fig. 9 listing in [1] also uses `sigmaC` and `t1_50year` on line 64 before lines 66–67 define them, and contains the typo `zeros(Nsim .1)`.

Correction. Compute the lifetime index of each event directly and accumulate in one vectorised step:

```
life = floor(earthquake_time/Nyears) + 1;                % 1..Nsim
dT   = (stress_eq/sigmaC).^n * t1_50year;                 % per event
T    = teT_baseline + accumarray(life, dT, [Nsim 1]);     % empty -> baseline
Ft   = 1 - exp(-(K*sigmaC.^n*T).^beta);
```

This version has no ordering assumption, handles empty lifetimes, and needs no sort.

C1 and C2 must be corrected together. The two defects partly cancel. Correcting C1 alone leaves about one event per lifetime, almost all of which the original loop discards. In the synthetic case of Section 3.5, correcting C1 alone lowers $\bar{F}$ from 1.0×10^{-5} to 1.0×10^{-7}.

3.3 C3 — Intensity sampling and upper truncation

Finding. The bins span $\log_{10}$ return periods 1.7–3.4 (50 to $2{,}512$ years). Table 3 lists the return periods to which `post_process_s91_stress.m` maps the scaled CMS runs: the SLE $\times 1.36$ and $\times 1.85$ runs fall at about $6{,}900$ and $21{,}600$ years and are never used with the mean hazard curve. Because equivalent time scales as $s^{30.5}$, rare large events can dominate the expectation even though they are infrequent. With synthetic statistics in which seismic stress dominates, extending the range to $10^{4.3}$ years raises $\bar{F}$ about 20 times (Table 4). Whether the highest return periods contribute is therefore a property of the data, to be checked rather than assumed.

Table 3: Return periods assigned to the scaled CMS runs (`rp_study`), recomputed from the $T = 0.1$ s hazard curve of [4] (Table 4-1) with OLE Sa $= 0.455$ g and SLE Sa $= 1.537$ g. The OLE $\times 1.85$ run is not used; its Sa (0.84 g) coincides with that of SLE $\times 0.54$ (0.83 g), and the two give the same peak force ($6{,}562$ vs $6{,}704$), which supports D. Schwartz's comment on why it was dropped.

Scale factor	0.54	0.735	1	1.36	1.85
OLE CMS: return period, mean hazard (yr)	28	52	100	203	—
SLE CMS: return period, mean hazard (yr)	425	977	$2{,}475$	$6{,}881$	$21{,}558$
OLE CMS: return period, Sa/1.3 (yr)	17	31	57	110	—
SLE CMS: return period, Sa/1.3 (yr)	219	480	$1{,}117$	$2{,}858$	$8{,}132$

Correction. Sample the intensity of each earthquake directly from the hazard curve. For a lower intensity bound $a_{\min}$, the events with $\mathrm{Sa} > a_{\min}$ form a Poisson process of rate $\lambda(a_{\min})$, and the intensity of each event has the conditional survival function

$$P(\mathrm{Sa} > a \mid \mathrm{Sa} > a_{\min}) = \frac{\lambda(a)}{\lambda(a_{\min})}. \tag{8}$$

Inverse-transform sampling then gives

$$N_\ell \sim \mathrm{Poisson}(50\,\lambda(a_{\min}))\,, \qquad a = \lambda^{-1}(u\,\lambda(a_{\min}))\,,\ u \sim \mathcal{U}(0,1), \tag{9}$$

per lifetime ℓ, with λ^{-1} evaluated by log–log interpolation of the hazard table. This removes the bins (and C1 with them) and uses the full hazard curve up to its last tabulated point (2.8 g, about $20{,}000$ years at $T = 0.1$ s). The response must still be known at those intensities: either the stripes of Section 4.1 reach that far, or the extrapolation beyond the largest run is stated and its sensitivity reported. In either case, report the convergence of $\bar{F}$ as the upper bound is raised. The lower bound $a_{\min}$ can be set where $F(T_b + t_{\mathrm{eq}}(a)) - F(T_b)$ is negligible; a 50-year return period, as in WE91, is a reasonable default to be checked the same way.

3.4 C4 — Reproducibility and housekeeping

1. **Random seed.** No seed is set, so results are not reproducible. In the synthetic cases $\bar{F}$ varies by 10–30% and $s_{\mathrm{eq}}^{\mathrm{MC}}$ by 4–8 psi from seed to seed. Set `rng(seed)`, report the standard error $\mathrm{std}(F_\ell)/\sqrt{N_{\mathrm{sim}}}$ alongside $\bar{F}$, and use common random numbers when comparing cases (mean versus fractile hazard, or different safety factors) so that their difference is not dominated by sampling noise.
2. **Safety-factor sweep.** Line 235 defines `sf_mechanical = 1:.2:5`, which line 236 overwrites with `2`. Figure 1002 therefore shows one point, and the block that plots `sf_mechanical = 3.5` never executes. Remove line 236 to recover the curves of Fig. 15 of [2].
3. **Thermal and annealing factor.** The code uses `sf_thermal_anneal = 1`; the SPIE paper [3] used 2.0. State which value is intended.
4. **Figure 301.** The legend labels are swapped, the abscissa uses Sa after the division by 1.3 (lines 201–202 run first), and `k3` is left over from the previous loop, so only stress group 6 is shown.
5. **Comments.** Lines 37–38 label runs 33–37 and 38–42 both as OLE (see Table 2). Lines 61 and 80 initialise `Ft` as a column and `Te` as a row.
6. **Downselection.** `stress_post_local_7` keeps only the samples where the actuator force exceeds 50% of its maximum. Because the equivalent time scales as $s^{30.5}$, the discarded samples contribute less than $0.5^{30.5} \approx 10^{-9}$ of the peak, so this is harmless. GMT-DOC-06175 should state it, since its description of Equation (1) assumes a regularly sampled series.

3.5 Effect of the implementation corrections

Table 4 compares the variants on two synthetic sets of equivalent-stress statistics (means in psi for the OLE scale factors 0.54–1.36 and the SLE scale factors 0.54–1.85, coefficients of variation 0.06 and 0.13, baseline $s_b = 300$ psi):

- case A, seismic-dominated: OLE $(230, 260, 300, 350)$, SLE $(380, 450, 540, 650, 800)$;
- case B, milder: OLE $(150, 170, 195, 225)$, SLE $(240, 280, 330, 390, 470)$.

Table 4: Effect of the implementation corrections on synthetic inputs. $\pm$ is the seed-to-seed standard deviation. The baseline-only probability is $F(T_b) = 7.4 \times 10^{-8}$.

		Case A		Case B	
Variant	Events	$\bar{F}$	$s_{\mathrm{eq}}^{\mathrm{MC}}$ (psi)	$\bar{F}$	$s_{\mathrm{eq}}^{\mathrm{MC}}$ (psi)
Original `.m` (as run in WE91)	21,187	3.4×10^{-5}	468 ± 4	1.08×10^{-7}	308.5
GMT-DOC-06175 Fig. 9 loop	21,187	4.4×10^{-5}	476 ± 4	1.18×10^{-7}	310.5
C2 corrected	21,187	4.4×10^{-5}	476 ± 4	1.18×10^{-7}	310.5
C1 + C2 corrected	$\approx$ 4,900	1.0×10^{-5}	428 ± 5	0.84×10^{-7}	302.8
C1 + C2, lognormal sampling	$\approx$ 4,900	1.2×10^{-5}	434 ± 8	0.87×10^{-7}	303.6
C1 + C2 + C3 (range to $10^{4.3}$ yr)	$\approx$ 5,000	2.1×10^{-4}	534 ± 8	2.5×10^{-7}	327.7
C1 corrected, original loop	$\approx$ 4,900	1.0×10^{-7}	—	—	—

Each value is the mean over 20 seeds with $N_{\mathrm{sim}} = 5000$.
The C3 row uses the synthetic statistics at the SLE $\times 1.36$ and $\times 1.85$ return periods (about 6,900 and 21,600 years), interpolated linearly in log return period. These are the intensities the WE91 mean-hazard run never reaches, so the size of this effect for the real mirror depends on the response measured there, which is why the stripes of Section 4.1 are needed.

4 Corrections to the method and inputs

4.1 M1 — The CMS anchor of the stress distributions

Finding. The method takes the mean of each Gaussian from a single CMS record and its dispersion from the 16 CS records. This is valid only if the CMS record responds like the centre of the CS ensemble. It does not:

- In GMT-DOC-06175 Fig. 4 (SLE), the CMS equivalent stress (red asterisk) lies beyond the largest of the 16 CS values in every one of the six stress groups.
- `S91res.mat` stores the peak loadspreader force of every run (Figure 2, Table 5). At SLE, the CMS record gives 1.97 times the CS mean, about 3.5 CS standard deviations above it. At OLE the ratio is 1.24.
- The two CMS records agree with each other: OLE $\times 1.85$ and SLE $\times 0.54$ are at almost the same Sa and give 6,562 and 6,704. The discrepancy is therefore between the 2018 CMS records and the 2019 CS records, not between OLE and SLE.

Table 5: CMS (scale factor 1) versus the CS ensemble at the same level, from `S91res.mat`. COV is the coefficient of variation of the 16 CS runs.

	SLE		OLE	
Metric	CS COV	CMS/CS mean	CS COV	CMS/CS mean
Peak displacement	0.27	1.86	0.12	1.26
Peak loadspreader force	0.28	1.97	0.11	1.24
RMS displacement	0.30	1.87	0.33	1.24
RMS loadspreader force	0.34	1.81	0.26	1.41

Why it matters. The 16 CS records are selected and scaled so that their spectra match the conditional spectrum, mean and dispersion, at the conditioning period [4]. Their response ensemble is the best available estimate of the response distribution at that hazard level. A mean anchored on an outlying record shifts every level of the distribution. In synthetic case A, lowering

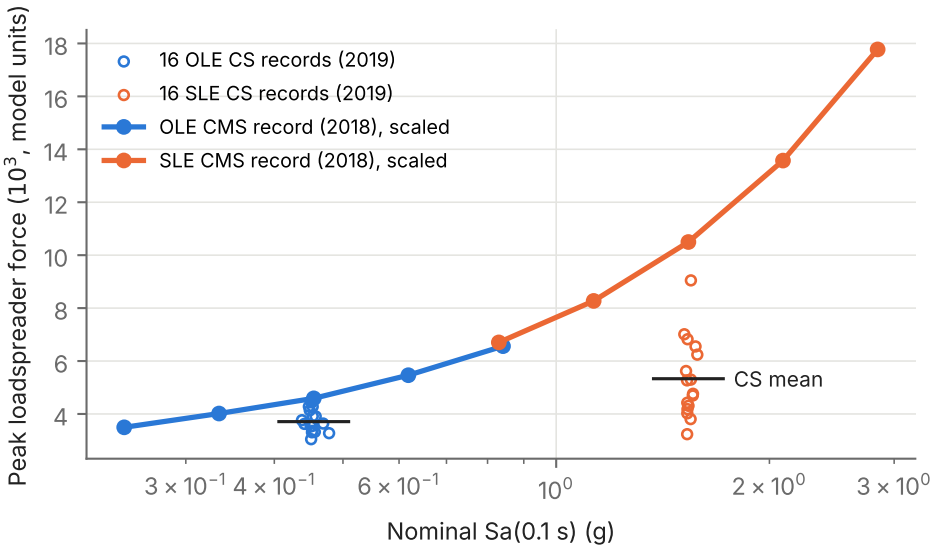


Figure 2: Peak loadspreader force for the 42 WE91 runs (`S91res.mat`, column 2) against the nominal Sa(0.1 s) of each run. CS records are jittered horizontally for legibility; the horizontal bars mark the CS means.

the SLE means by 15% lowers $\bar{F}$ by a factor of 8 and $s_{\mathrm{eq}}^{\mathrm{MC}}$ by about 60 psi; this is the largest single effect found in this review.

Probable cause, to be confirmed. AECOM computed the hazard for Vs30 = 760 m/s and converted it to the site's Vs30 = 1500 m/s with a constant factor of 0.7 on spectral acceleration, derived in the 2018 addendum [4]. If the 2018 CMS records were matched to the 760 m/s spectrum, their Sa(0.1 s) would be $1/0.7 = 1.43$ times too high. Given the sub-linear response of the isolated system (scaling the OLE CMS by 1.36 raises the force by 1.19), this would give a force ratio of about 1.25 at OLE, as observed. It does not explain the SLE ratio as well, so the hypothesis must be tested.

Corrections. In increasing order of effort:

1. **Diagnostic.** Compute the 5%-damped Sa(0.1 s) of the horizontal components of `SLE_short_2018_1`, `OLE_short_2018_1` and the 32 `*_2019_*` records, and compare with the targets of [4] Tables 2-3 and 2-4 (0.455 g and 1.537 g mean UHS at Vs30 = 1500 m/s).
2. **Map from measured intensity.** Assign each CMS run its return period from its *measured* Sa(0.1 s) times its scale factor, instead of the nominal `oleSa` $\times \{0.54, \dots\}$. This is correct whatever the cause of the discrepancy and costs nothing.
3. **Level from the CS, trend from the CMS.** Let $g(a)$ be the response of the scaled CMS runs as a function of Sa. The scaled runs are needed because the isolators and M1 dampers make the response nonlinear in intensity (D. Schwartz's comment in [1]). Anchor the level on the CS ensemble at each hazard level $L \in \{\mathrm{OLE}, \mathrm{SLE}\}$:

$$\tilde{s}(a) = \tilde{s}_{\mathrm{CS},L} \, \frac{g(a)}{g(a_L)}, \tag{10}$$

where $\tilde{s}_{\mathrm{CS},L}$ is the median of the 16 CS equivalent stresses at level L. Between the two levels, blend the OLE- and SLE-anchored curves linearly in $\log a$, as suggested in the review comments.

4. **Lognormal dispersion.** Model the equivalent stress at a given intensity as lognormal, with median $\tilde{s}(a)$ and log standard deviation $\sigma_{\ln}$ estimated from the 16 CS records at each level. Ground-motion intensity measures and structural responses are customarily lognormal; the conditional spectra themselves are defined in log space; and a lognormal cannot produce the negative stresses that a Gaussian can (which would turn $s^{30.5}$ complex in MATLAB). Report the sampling uncertainty of $\sigma_{\ln}$ from 16 samples: its relative standard error is about $1/\sqrt{2 \times 15} = 18\%$.

5. **Multiple-stripe analysis (new runs).** Scale all 16 CS records of each level to 4–6 intensities, spanning roughly 50 to $20{,}000$ years, and run each one. Fit the lognormal parameters at each stripe directly and drop the CMS anchor. This is the standard approach for nonlinear structures. It also provides the response at the rare intensities needed by C3, and it lets the Monte Carlo resample whole records (Section 4.3). Moderate scale factors applied to a record set near its own hazard level are standard practice; for large factors, check the scaled spectra against the conditional spectrum at the target level.

4.2 M2 — Long-term (baseline) stress

Finding. Equation (4) applies $s_b = 200 + SF \times 50$ psi to every stress group: `base_mech = 50` is described as the "assumed long term mechanical stress". The gravity stresses in the damper peer review [5], for segment 1, are listed in Table 6. At the backplate near the vent hole, the group identified as driving the static-support design [6], the gravity stress is about twice the assumed value.

Table 6: Gravity stress of segment 1 (global + local, psi), from [5].

Stress group	ZA 60°	ZA 0°
Faceplate	33.9	27.4
Web ends	22.7	17.8
Fillet, back plate	45.4	67.2
Cross-pin holes	38.5	38.9
Backplate near vent hole	99.7	110.6
All else	24.2	27.7

Why it matters. The baseline acts for the full 50 years, so for small F its contribution scales as $s_b^{n\beta} = s_b^{13.8}$. At $SF = 2$, the vent-hole baseline is $200 + 2 \times 110 = 420$ psi rather than 300 psi, which raises the baseline failure probability by $(420/300)^{13.8} \approx 100$. In synthetic case A this makes the baseline comparable to the seismic contribution, where with 50 psi it was negligible. The 50 psi value underestimates the vent-hole and fillet groups and overestimates the faceplate and web groups.

Correction. Compute the baseline per segment s and stress group g from a pure-gravity load case run through `Stress_upd_2025` (the inertial influence matrices A_i with 1 g applied at each zenith angle z [6]). Weight the zenith angles in *equivalent time*, not in stress:

$$T_{b,sg} = t_{50} \sum_z w_z \left(\frac{SF\, s_{sgz}^{\mathrm{grav}} + s_{\mathrm{th}} + s_{\mathrm{an}}}{s_r} \right)^n, \qquad \sum_z w_z = 1, \tag{11}$$

where w_z is the fraction of the lifetime spent at zenith angle z. The commented-out `weights = [7,3,2]/12` in `post_process_s91_stress.m` is the natural place for this. Weighting the stress instead would underestimate the baseline, because with $n \approx 30$ the worst zenith angle dominates.

Two consistency checks are needed:

- If the seismic stress series start from a gravity-initialised state (the workshop describes initialising from a gravity run [2]), the static stress of each group can be read from the pre-event samples *before* downselection. The baseline and the seismic contributions then come from the same model and the same elements.
- The handoff memo [6] questions whether the 100 psi thermal stress double-counts the stress from the glass–mount CTE mismatch. Settle this at the same time.

4.3 M3 — System failure probability and Weibull constants

4.3.1 Derivation of K and β

Both [2] (§1.7) and [1] (question 3) note that the origin of K and β is undocumented. They follow from the time- and area-dependent Weibull law fitted by the University of Arizona to witness samples [5]. For a surface of area A held at stress σ for time t,

$$-\ln(1-F) = \frac{A}{A_T}\left(\frac{\sigma^n\, t}{\sigma_C^n\, t_T}\right)^{m/n}, \tag{12}$$

where σ_C is the characteristic strength measured over the test area A_T and test time t_T, and m is the Weibull modulus. The term $\sigma^n t$ is the stress-time invariant of sub-critical crack growth, so Equation (12) can be written in the form of Equation (3) with

$$\beta = \frac{m}{n}, \qquad K = \frac{(A/A_T)^{n/m}}{\sigma_C^n\, t_T}. \tag{13}$$

With the Fibrefrax-mould parameters ($\sigma_C = 25.83$ MPa, $m = 13.775$, $n = 30.46$, $A_T = 1710$ mm^2, $t_T = 10$ s) and the stressed area $A = 2.83 \times 10^7$ mm^2 of [5], Equation (13) gives $\beta = 0.4522$ and $K = 3.00 \times 10^{-101}$ (psi, s), the values used in the code. Table 7 applies Equation (13) to the other surface finishes. The 1%, 50-year stress of each finish reproduces the group limits hard-coded in `post_process_s91_stress.m` (`grouplimits = [714,714,714,472,714,427]`), which confirms the derivation.

Table 7: Weibull constants per surface finish from Equation (13), $n = 30.46$, $A = 2.83 \times 10^7$ mm^2. The last column is the constant stress giving $F = 1\%$ in 50 years.

Finish	σ_C (MPa)	m	β	K (psi^{-n} s^{-1})	$s_{50}^{1\%}$ (psi)
Fibrefrax mould	25.83	13.775	0.4522	3.00×10^{-101}	713
QF-150 mould (cross-pin holes)	19.36	12.286	0.4033	2.64×10^{-96}	471
GMT1 repaired back, < 51 mm[a]	17.525	12.286	0.4033	5.48×10^{-95}	427
GMT1 repaired back, > 51 mm[a]	20.123	12.286	0.4033	8.13×10^{-97}	490

[a] Distance from the puck or vent hole.

The code uses the Fibrefrax constants for all groups, including the QF-150 cross-pin holes and the GMT1 back. It also uses $n = 30.5$ rather than 30.46; the difference is immaterial but should be made consistent.

4.3.2 Finding

Line 261 of `post_process_s91_stress.m` takes the maximum equivalent stress over the seven segments of each run, and the Monte Carlo then treats that value as a single mirror. Each stress group is evaluated separately, each with the full segment area $A = 2.83 \times 10^7$ mm^2 applied to the group's peak stress.

4.3.3 Correction: sum of risks of rupture

Crack initiation in Equation (12) is a weakest-link process on independent flaw populations. Given the stress history, flaws in different segments, and in disjoint regions of one segment, are independent, so the survival probabilities multiply. For lifetime ℓ,

$$F_\ell = 1 - \exp\Big(- \sum_{s=1}^{7} \sum_{g} R_{sg\ell} \Big), \qquad R_{sg\ell} = (K_g \, s_r^n \, T_{sg\ell})^{\beta_g} \, , \tag{14}$$

where $T_{sg\ell}$ is the accumulated equivalent time of group g on segment s, and K_g, β_g come from Equation (13) with the area A_g of group g and the parameters of its finish. Three points follow:

- Taking the maximum over segments discards the other six terms of the sum. When the segments are similarly loaded, the system probability is up to 7 times the maximum-segment probability.
- Once the terms are summed, the group areas A_g (from the element groups in `stress_groups.neu` and `stress_groups_CS.neu`) must replace the full-segment area. Keeping $A = 2.83 \times 10^7$ mm^2 for every group and summing would count the mirror area several times.
- Applying the peak stress of a group to its whole area remains conservative. If that conservatism is too large, element-level risk integration, $R = \sum_e (A_e/A_T)\,(\cdots)^{m/n}$, is the exact form of Equation (12).

4.3.4 Correction: resample records, not parameters

Sampling an independent Gaussian for each quantity would lose the correlation between segments and groups: one earthquake loads all segments together. Resampling whole records keeps it. For each simulated earthquake, draw one record index r (and one orientation, Section 4.5) and use that record's complete 7×6 array of equivalent times, interpolated to the event's intensity. With multiple stripes, the interpolation is done per record in $\log t_{\mathrm{eq}}$ versus $\log a$ between stripes. With the current data, it can be approximated by scaling each CS record's array with the trend of Equation (10). This non-parametric bootstrap also removes the need to assume any distribution shape, which [2] notes 16 runs cannot confirm.

4.4 M4 — Epistemic hazard uncertainty

Finding. The "fractile" case divides the OLE and SLE spectral accelerations by 1.3 before mapping them to return periods, which is equivalent to stretching the mean hazard curve by 1.3 along the Sa axis. The code and the workshop document [2] call this the 95th-percentile hazard; GMT-DOC-06175 attributes it to modelling uncertainty. AECOM [4] (Tables 5-1 and 5-2) gives the fractile spectra at the conditioning period (Table 8). The 95th-percentile to mean ratio is 1.22 at OLE and 1.16 at SLE, so the factor 1.3 is conservative relative to the stated percentile.

Table 8: Fractile UHS at $T = 0.1$ s (Vs30 = 760 m/s, g), from [4]. The ratios to the mean do not depend on the 0.7 site factor.

Level	5th	15th	50th	mean	85th	95th	95th/mean
OLE (100 yr)	0.479	0.519	0.649	0.652	0.756	0.797	1.22
SLE (2475 yr)	1.545	1.690	2.261	2.199	2.468	2.561	1.16

Corrections.

1. **Now.** Replace the factor 1.3 with the ratio $\rho(x)$ interpolated linearly in log return period x between 1.22 at 100 years and 1.16 at 2,475 years, held constant outside that range. Label the case "95th-percentile hazard (epistemic)".
2. **Proper treatment.** Request the fractile hazard *curves* at $T = 0.1$ s from AECOM (5th, 15th, 50th, 85th, 95th). Addendum 2 gives fractile spectra only at the OLE and SLE levels. Run the Monte Carlo once per fractile curve, so that the epistemic uncertainty in the hazard is reported as a distribution of $\bar{F}$. The $\bar{F}$ obtained with the 95th-percentile curve is the failure probability conditional on that curve, not the 95th percentile of the failure probability.
3. Keep the modelling uncertainty (damping, nonlinearities, stress model, analyst choices) as a separate term, as the SPIE paper does [3], instead of folding it into the hazard factor.

4.5 M5 — Stress model and telescope orientation

Finding. WE91 uses the `Stress_upd` model. The handoff memo [6] reports a large discrepancy between that model and the University of Arizona analyses, and replaces it with `Stress_upd_2025`, built on UA-generated influence matrices. The off-axis segment model has been checked against UA; the centre-segment model has not. All 42 runs are at azimuth $0°$ and zenith angle $60°$; the workshop notes that a proper fragility analysis would include more records and more elevation and azimuth angles [2].

Corrections.

1. Re-post-process all runs with `Stress_upd_2025`. Flag the centre-segment (segment 7) results until its validation is complete.
2. Treat zenith and azimuth angles as aleatory variables. Run zenith angles $0°$, $30°$ and $60°$, and in the Monte Carlo draw the zenith angle of each event with the weights w_z of Equation (11).
3. If the base and mount are approximately symmetric in azimuth, rotating the two horizontal ground-motion components by a random angle is an inexpensive substitute for azimuth runs. Verify this with one run at a second azimuth before relying on it.
4. Approximate budget: 16 records $\times$ 5 stripes $\times$ 3 zenith angles ≈ 240 simulations.

5 Revised algorithm

Algorithm 1 combines the corrections. The inputs are the equivalent-time array $t_{\mathrm{eq}}[r, z, k, s, g]$ for record r, zenith angle z, stripe k (intensity a_k), segment s and group g, computed from the mechanical stress multiplied by SF plus the thermal and annealing stress; the baseline $T_{b,sg}$ of Equation (11); the per-group constants K_g, β_g; and the hazard curve $\lambda(a)$.

Each lifetime is simulated independently, so the lifetime index is exact and empty lifetimes keep their baseline probability (C2). Repeat the whole algorithm for each fractile hazard curve (M4) and each safety factor, with common random numbers. When reporting a single equivalent stress for a system that mixes finishes, also report $\bar{F}$ itself, because Equation (5) is defined for one set of Weibull constants.

6 Verification

The following tests should pass before any result is used.

1. **No earthquakes.** With $\lambda \equiv 0$, $\bar{F} = 1 - \exp(-\sum_{s,g}(K_g s_r^n T_{b,sg})^{\beta_g})$ exactly.

Algorithm 1: Corrected M1 fragility Monte Carlo.

Input: $t_{\mathrm{eq}}[r, z, k, s, g]$, a_k, $T_{b,sg}$, K_g, β_g, $\lambda(\cdot)$, w_z, $a_{\min}$, N_{sim}, seed
Output: $\bar{F}$, its standard error, $s_{\mathrm{eq}}^{\mathrm{MC}}$

initialise the random generator with the seed
for $\ell \leftarrow 1$ **to** N_{sim} **do**
  $T_{sg} \leftarrow T_{b,sg}$ `# baseline, M2`
  $N \sim \mathrm{Poisson}\big(50\,\lambda(a_{\min})\big)$ `# event count, C1`
  for $j \leftarrow 1$ **to** N **do**
    $u \sim \mathcal{U}(0,1)$; $a \leftarrow \lambda^{-1}\big(u\,\lambda(a_{\min})\big)$ `# intensity, C3`
    $r \sim \mathcal{U}\{1,\dots,16\}$; $z \sim w_z$ `# record and orientation, M3 and M5`
    $T_{sg} \leftarrow T_{sg} + \exp\Big(\mathrm{interp}_{\log a}\big(\log a_k,\ \log t_{\mathrm{eq}}[r,z,k,s,g],\ \log a\big)\Big)$
  $F_\ell \leftarrow 1 - \exp\big(-\sum_{s,g}(K_g\, s_r^n\, T_{sg})^{\beta_g}\big)$ `# system probability, M3`
$\bar{F} \leftarrow \mathrm{mean}(F_\ell)$; $\mathrm{SE} \leftarrow \mathrm{std}(F_\ell)/\sqrt{N_{\mathrm{sim}}}$
$s_{\mathrm{eq}}^{\mathrm{MC}} \leftarrow$ Equation (5) with the constants of the governing finish

2. **Event statistics.** Over N_Y years, the number of simulated events with return period at least X is Poisson with mean N_Y/X (the check described in [3]). The original code fails this test by a factor of 4.3.

3. **First-order analytic check.** When lifetimes rarely contain more than one damaging earthquake (D. Schwartz's remark in [1]), the expectation reduces to a hazard convolution:

$$\bar{F} \approx F(T_b) + 50 \int_{a_{\min}}^{\infty} \mathbb{E}\big[F\big(T_b + t_{\mathrm{eq}}(a)\big) - F(T_b)\big]\,\big|d\lambda(a)\big|. \tag{15}$$

   For synthetic case A, Equation (15) gives 1.063×10^{-5} and the corrected Monte Carlo gives $(1.009 \pm 0.033) \times 10^{-5}$ ($10 \times 20{,}000$ lifetimes). Equation (15) slightly overestimates because F is concave in T when several events share a lifetime. Agreement within a few percent is the expected outcome, and Equation (15) is an inexpensive, independent screening tool in its own right.

4. **Convergence.** $\bar{F}$ is stable to within its standard error when N_{sim} is doubled, when $a_{\min}$ is lowered, and when the upper limit of the hazard curve is raised.

5. **Regression.** With C1 and C2 re-introduced as options, the new implementation reproduces the original WE91 numbers within seed noise.

7 Implementation plan

Table 9 orders the work so that the corrections that need no new simulations come first. Steps 1–4 can be completed with the existing S91 data. Step 3 is self-contained and suitable as a first module of an independent re-implementation of the seismic tool.

8 Responses to the questions in GMT-DOC-06175

1. *Is more CS ground-motion data available?* No; AECOM supplied as many records as were available [1]. Multiple-stripe scaling of the existing 32 records (M1) and orientation sampling (M5) make better use of them, and the record bootstrap (M3) avoids assuming a distribution that 16 samples cannot confirm.

Table 9: Prioritised implementation plan.

Step	Work	Inputs	New runs
1	Sa(0.1 s) check of the 2018 and 2019 records (M1, diagnostic); remap CMS runs by measured Sa.	Time series in `SeismicTimeSeries`; [4] Tables 2-3, 2-4.	No
2	Per-segment, per-group gravity baseline (M2); per-group K_g, β_g and areas (M3).	`Stress_upd_2025` inertial matrices; `stress_groups*.neu`; Table 7.	No
3	Restructured Monte Carlo (Algorithm 1): C1–C4, M3 bootstrap, verification tests of Section 6.	S91 stress files (`T:\SeismicResults\S91`).	No
4	AECOM fractile ratios (M4, now); request full fractile hazard curves.	[4] Tables 5-1, 5-2.	No
5	Multiple-stripe runs at three zenith angles with `Stress_upd_2025` (M1, M5, C3).	32 CS records; seismic tool.	≈ 240

2. *Are RLE time series needed?* At the frequent, low-intensity end, the contribution is negligible because $t_{\text{eq}} \propto s^{30.5}$, in agreement with D. Schwartz's reply. The gap that matters is at the rare, high-intensity end (C3), which WE91 truncates at $2{,}512$ years.
3. *Is there a reference for n, K and β?* $n = 30.46$ and the Weibull parameters come from the UA witness-sample tests summarised in [5]; K and β follow from Equation (13) and are reproduced in Table 7.
4. *Should the safety factor apply to the random seismic contribution?* It already does. Line 254 of `post_process_s91_stress.m` multiplies the full mechanical stress series (seismic and gravity) by `sf_mechanical` before the equivalent-stress statistics are computed, and `sf_mechanical` also multiplies `base_mech` in the baseline. The safety-factor sweep is disabled by line 236 (Section 3.4).

A Reference implementation of the corrected bookkeeping

The Python functions below are taken from `mc.py`, which also contains the original loop and the GMT-DOC-06175 Fig. 9 loop for regression testing.

```python
import numpy as np
K, beta, n = 3e-101, 0.4522, 30.5
sC, t50 = 250.0, 3600*24*365*50
pf = lambda T: 1 - np.exp(-(K*sC**n*T)**beta)

def event_counts(rp_bins, n_years, rng):
    # C1: Poisson counts from the difference of exceedance rates at bin edges
    lam = 10.0**(-np.asarray(rp_bins))
    return rng.poisson(n_years*(lam[:-1] - lam[1:]))

def lifetimes(t, s_eq, n_sim, s_base):
    # C2: exact lifetime index; empty lifetimes keep the baseline
    life = np.floor(t/50).astype(int)
    T = (s_base/sC)**n*t50 + np.bincount(life, weights=(s_eq/sC)**n*t50,
                                         minlength=n_sim)
    return pf(T)
```

References

[1] R. Romano, *M1 Stress Fragility Analysis: a low-level description of the method implementation*, GMT-DOC-06175, v1.0 draft, GMTO, Feb. 2025 (with comments by D. Schwartz, Apr. 2025).

[2] D. Schwartz, *GMT Seismic Workshop*, GMT-XXX-XXXXX, Ver. 1 draft, GMTO, Jan. 2023.

[3] D. Schwartz, T. Ranka and D. Ashby, "Protecting a fragile giant: seismic design, analysis, results and requirements for the GMT," *Proc. SPIE* 11450, 114500S, 2020.

[4] C. B. Crouse, *Addendum 2 to Final Report on Update of AECOM Site-Specific Seismic Hazard Analysis for GMT Site*, AECOM-60613786, Jan. 2020.

[5] D. Schwartz, *M1S SA Damper Peer Review, Section 1: Damper Analysis*, GMTO, slides.

[6] D. Schwartz, *Seismic Analysis Handoff*, V1, GMTO, 15 Jan. 2026.

[7] D. Schwartz, *GMT System Level Analysis: Seismic Acceleration and Seismic Isolator Modeling*, GMT-DOC-03788, Mar. 2019.